

A Driver-Centric SoC and Geospatial Distance-Based Intelligent System for EV Charging Station Assistance

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Abstract. The escalating adoption of electric vehicles calls for intelligent and reliable charging infrastructure to address challenges such as optimal battery utilization and range anxiety. Existing electric vehicle charging station recommendation techniques rely on optimization-based formulations. These methods require careful parameter tuning, or take into account the distance, travel time, and cost, often ignore vehicle's state of charge as a critical factor. Furthermore, Haversine-based distance estimation, a geospatial technique, has not been widely utilized in EV charging applications. This paper proposes a lightweight, driver-centric, and SoC-aware intelligent assistant framework to recommend the top 4 nearest charging stations. The proposed model integrates predicted SoC, real-time geo-location, and journey distance. The system employs modular architecture comprising a back end, database, and front-end. The database stores static charging station details along with SoC histories and dynamic travel, which are pre-processed to support accurate modeling. The back end incorporates a Haversine-based distance computation module and a XGBoost model –based SoC predictor. Observations show that model performs comparatively with lower mean square error of 27.7796, root mean square error of 5.2706, mean absolute error of 1.9054, and R^2 of 0.9324. The front end provides an interactive platform displaying four nearest charging stations along with the predicted SoC upon arrival. The real-time evaluations are conducted across locations in the Kerala and Kerala–Tamil Nadu border due to adequate availability of charging station details. The observations demonstrate the effectiveness of the proposed framework in delivering accurate reliable distance computation, SoC prediction, and timely EVCS recommendations. While the validation is geographically constrained, the results confirm the viability of the proposed module as a practical solution to reduce range anxiety and enhance EV user convenience.

Keywords: Electric Vehicle Charging Station, Geo-location, Haversine Formula, Prediction, Random Forest, State of Charge.

1 Introduction

The global transportation sector is undergoing a paradigm shift from conventional internal combustion (IC) engine [1 – 2] vehicles to electric vehicles (EVs), driven by

the urgent need to improve energy efficiency, reduce greenhouse gas emissions [3], and minimize dependence on fossil fuels. Globally, EV adoption is accelerating at an unprecedented rate [4 – 5]. From a niche segment just half a decade ago to representing over one in five new car sales today, EVs are swiftly transforming transportation. Batteries that replace the fuel tank of IC engine vehicles serve as the main energy source in EVs. The Battery Management System (BMS) is an inevitable component in EVs to ensure efficient, safe, reliable operation of the battery pack. It continuously tracks key parameters [6] such as current, voltage, State of Charge (SoC) and temperature. Precise SoC estimation allows control systems and drivers to foresee the usable driving range and plan charging activities accordingly. [7] explores users' EV charging location preferences among home, gas stations, work, shopping malls, grocery stores, and other locations in the Okanagan region of British Columbia. [8] presents a model that guides EVs to optimal electric vehicle charging station (EVCS) based on forecasted load profile, drone-assisted edge computing and optimal path planning. [9] deploys explainable AI and Machine Learning algorithms to categorize EVCS. The authors have proved better performance of Decision Tree over KNN, Logistic Regression, SVM, and Random Forest Classifiers. [10] proposes an application to track the vehicle operation and notify the driver to search for EVCS. [11] employs machine learning approach along with the Shapley Additive Explanations method to model the charging decisions of EV users. [12] put forward a real-time method to recommend EVCS based on minimal current information like distance, time, and fees, while driving to the destination. [13] develops an algorithm to recommend EVCS and the associated energy prices by deploying an optimization problem. [14] presents an immune algorithm to locate the optimal charging station that analyses and process real time data from BMS. A deep reinforcement learning based fast charging station recommendation system is proposed in [15]. [16] deploys LightGBM to predict user ratings based on charging orders. [17] suggests an EVCS recommendation model considering the latitude and longitude of the charge point, vehicle type, and average speed of the vehicle. [18] develops a model to identify the nearest charging station by using the geodesic distance and station index. The geodesic distance is computed based on current location of vehicle and EVCS location. [19] proposes the allocation of medicine dispensing Robot based on the battery SoC and optimal path. However, the method in [20] accepts the current SoC for identifying the optimal path. [21] identifies the best SoC predictor among random forest, linear regression, XGBoost and CatBoost models. [22] utilizes CV data to compute trip lengths associated with EV, ICE or Hybrid Vehicle.

Existing EVCS recommendation approaches predominantly derive solutions through the formulation of optimization or objective functions [13 – 15]. However, these methods require careful mathematical formulation and parameter tuning to achieve convergence and optimality. In contrast, machine learning (ML)-based techniques rely on large datasets to learn underlying patterns to deliver accurate recommendations, while eliminating the need of explicit objective-function. Several studies use ML classifiers to select charging stations [9, 17]. A few methods consider travel distance, time, or charging cost as decision parameters [10, 12, 17]. However, these frameworks typically omit EV's SoC as a decisive factor in station selection. Moreo-

ver, most of these systems fail to provide users with the predicted SoC upon arrival at different EVCS locations, the critical information to alleviate range anxiety and enable decision-making. Geospatial computation techniques have also been incorporated in some models [17–18], where distance based on latitude and longitude is used to determine proximity between charging stations and EV. Studies in [23 – 24] specifically deploy the Haversine formula to compute the great-circle distance between coordinates against destination and source. However, this approach has not yet been extensively utilized within electric vehicle (EV)–related research, such as EVCS recommendation or SoC prediction frameworks. Despite these advancements, limited research has addressed a driver-centric, lightweight, and SoC-aware assistive framework capable of seamlessly integrating with existing EV onboard systems while providing accurate, real-time charging recommendations. Furthermore, most existing studies do not perform a comparative analysis of multiple SoC prediction models to identify the most accurate one for EV energy estimation. In addition, current recommendation systems fail to generate and rank the top nearest EVCS locations based on geospatial distance and predicted SoC, which restricts their practical applicability in real-time navigation and charging decisions.

The paper aims to develop an intelligent **lightweight** assistant framework for EV users to provide real-time notifications on four nearest EVCS, based on the travel distance computed from vehicle’s and EVCS geo-locations, current SoC, speed and time. The model also intimates the users with the remaining SoC upon arriving at the destination. The proposed system integrates machine learning models with geo-spatial data to predict battery performance and recommend reachable EV charging stations, thereby minimizing the risk of battery depletion and reducing range anxiety. By leveraging location-based services, distance estimation, and predictive analytics, intelligent assistance enhances route planning, ensures energy-efficient travel, and improves the overall reliability of EV adoption in smart transportation systems.

2 Proposed Model

The paper aims to develop an intelligent model that recommends four nearest EVCS along with the SoC after arriving at the destination. The system incorporates a user-informative front end and a backend. The front end has the provision to accept the current geo location of EV and display the four nearest EVCS along with the battery remaining SoC after arriving at the destination. The schematic of the framework is shown in Fig. 1.

The Front end (shown in Fig. 1) is the user-facing layer, interacting with the other layers and providing information to the driver. It accepts the current EV Geo-location and displays the list of nearby EVCS to the driver, usually filtered based on availability, distance, or SoC needs along with the SoC consumed post travelling to the destination. The back end is the processing and computational layer that powers the system. It includes a database of EV charging station locations (Geo-coordinates, location, etc.), distance calculator and SoC predictor model. The distance calculator calculates the distance between the EV’s current position and potential EVCS locations

using Haversine Formula. Haversine formulas are commonly used in navigation. The calculation given in equations (1) – (5) is based on haversine formula uses the difference in latitude and longitude of source and destination.

$$a = \sin^2(\Delta\phi \div 2) + \cos(\phi_1)\cos(\phi_2)\sin^2(\Delta\lambda \div 2) \quad (1)$$

$$c = 2 \cdot \arctan2(\sqrt{a}, \sqrt{1-a}) \quad (2)$$

$$d = r \cdot c \quad (3)$$

Where,

d is the distance between the two points (in km or m)

r is the earth's radius (≈ 6371 km)

ϕ_1 and ϕ_2 are the latitudes of the source and destination

λ_1 and λ_2 are the longitudes of the source and destination

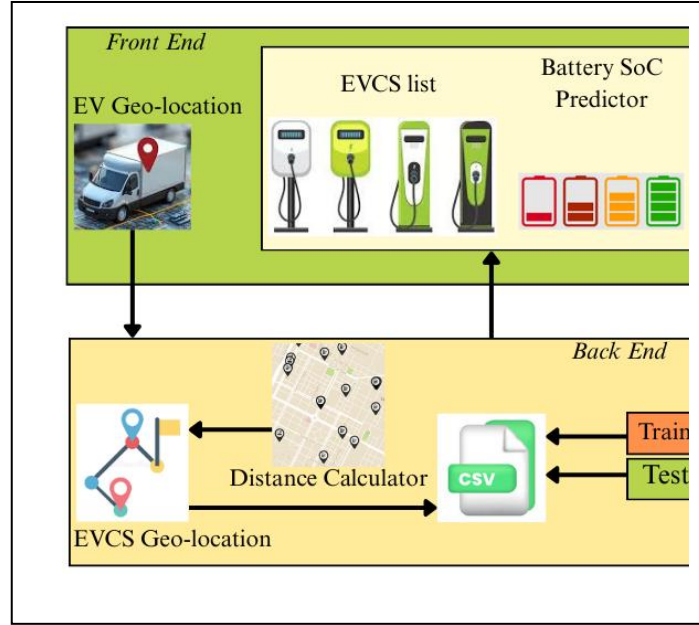


Fig. 1. Schematic of the proposed EVCS recommendation system.

$$\Delta\phi = \phi_2 - \phi_1 \quad (4)$$

$$\Delta\lambda = \lambda_2 - \lambda_1 \quad (5)$$

The machine learning based SoC prediction model is trained using historical datasets, available in CSV format. The model displays the SoC required to reach the destination. This ensures the recommendation of only reachable charging stations. The front end then presents the EVCS list along with battery SoC prediction to the user, guiding them to the most suitable charging station. This system combines geo-location data, distance calculation, and battery SoC prediction to recommend the most suitable EV

charging stations in real time, enhancing convenience and preventing battery depletion during travel. The feasible charging stations based on travel distance and consumption can be selected by the user.

3 Dataset Exploration

The paper deploys two datasets that include details of EVCS in India and battery parameter values during discharging. The EVCS dataset is taken from Kaggle [25] that is created using live data pulled from the OpenChargeMap API. The dataset contains over 480 EVCS details across 18 different states in India. This includes 436, 32, 4, 4, 3, 2, 1, 1, 1, 1, 1 station across Kerala, Tamil Nadu, Maharashtra, West Bengal, Karnataka, Uttar Pradesh, Telangana, Delhi, Madhya Pradesh, Andra Pradesh, Puducherry respectively. The data covers EVCS details under 10 operators, across 281 cities. The state details of 1 entry had missing entries and hence been removed. The details include Station Name, city, state, latitude, longitude, operator name, usage type (public, membership, etc.), connector type (CCS, Type 2, etc.) and power output (kW). The dataset is deployed to find the geo-locations of EVCS. This information, stored in the back end of the framework, is deployed to find the distance between the EVCS and EV current location. Table 1 presents details of seven EVCS. The Battery Soc dataset obtained from Kaggle [26] is intended to predict the SoC consumption of an EV when traveling from a source location to a destination. The dataset includes the source latitude and longitude, destination latitude and longitude, speed of the vehicle during the trip, distance between source and destination, altimetry values representing the up-and-down elevation profile of the route, estimated travel time in minutes and battery consumption required to reach the destination.

Table 1. Sample of EVCS dataset.

Station Name	City	State	Latitude	Longitude	Operator	Usage Type	Connector Type	Power (kW)
Tata.ev Luxon Motors - Tata Power	Edappally	Kerala	10.01148	76.31139	Tata Power	Public - Membership Required	CCS (Type 2)	60
Tata.ev Gokulam Motors - Tata Power	Choornikkara	Kerala	10.0813	76.33946	Tata Power	Public - Membership Required	CCS (Type 2)	60
Taj Cochin Inter- nation Airport Hotel - Tata Power	Nedumbassery	Kerala	10.16	76.39015	Tata Power	Public - Membership Required	CCS (Type 2)	30
Luxon Motors - Tata Power	Kariyad	Kerala	10.16204	76.36736	Tata Power	Public - Membership Required	CCS (Type 2)	30
Sree Gokulam Motors - Tata	Angamaly	Kerala	10.22106	76.37675	Tata Power	Public - Membership	CCS (Type 2)	30

The Kaggle Battery SoC dataset provides valuable insights for modeling EV state of charge and route characteristics; however, it has several limitations. First, the dataset covers a limited set of geographic routes, which can restrict the generalizability of trained models to other regions or road conditions. Second, vehicle heterogeneities like capacity, differences in battery chemistry, and weight, are not fully represented, which affects the model performance. Third, environmental, long-term battery degradation and temporal factors like weather, ambient temperature, traffic conditions, and are not captured. These factors can influence SoC dynamics and energy consumption. In addition, dynamic driving behaviors such as deceleration, acceleration, speed variability and sufficient resolution of altimetry data to fully represent road gradients, are not represented. These limitations highlight the need for cautious interpretation of model results. The paper considers the vehicle speed, Geo-location of EVCS, and travel time as the input features. The output is the SoC consumption required for the EV to reach the selected charging station.

The collected raw datasets often contain redundant, inconsistent, or missing values which can significantly reduce the model's accuracy. Since EVCS recommendation and SoC prediction rely greatly on consistent temporal alignment between variables and numerical precision, preprocessing becomes an essential step. Careful preprocessing of data ensures learning of model from clean, standardized samples. Hence, the dataset is pre-processed to remove NaN and missing values. The observations with NaN in any of the features are removed. Meanwhile, rows with missing values are removed from the dataset. The values are then normalized using standard mix-max method to bring down the feature range $[0, 1]$. The paper considers Random Forest and XGBoost, that have demonstrated superior performance in many applications [27 – 29] for SoC prediction. The best model is selected for the proposed system.

4 Results and Discussions

The goal of the paper is to create an intelligent assistant framework for EV travellers that, based on the vehicle's geolocation, current battery SoC, and vehicle speed, can deliver real-time notifications on the best charging stations. The proposed assistant for EV travelers is designed with a modular architecture, broadly categorized into the front end, back end, and database layers. Each module plays a distinct role in enabling real-time EVCS recommendation and battery SoC prediction. The database stores historical SoC, EV charging station details, and travel data, and user-specific charging and driving patterns. This supports accurate model prediction and personal recommendations. The back end performs computations that include distance, ML based SoC prediction, and charging station recommendation based on battery status and geo-location.

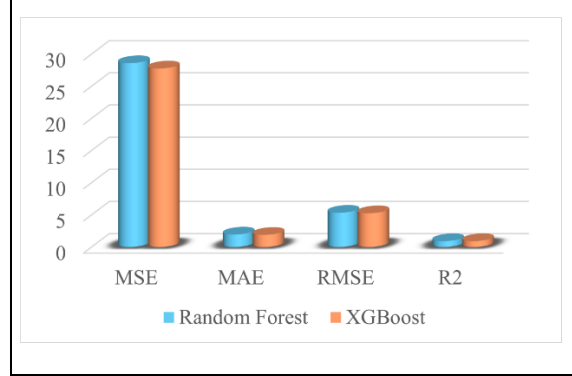


Fig. 2. Comparison of performance indices – Random Forest and XGBoost.

The front end provides an interactive platform that accepts the geo-location of EV and displays the four nearest charging stations along with the remaining SoC upon reaching each station. This assists travelers in making informed charging decisions. The database serves as the knowledge repository for the system, storing both static and dynamic data, essential for accurate forecasting and recommendations. The comprehensive EVCS information such as station name, city, state, operator name, connector type and power output are collected as mentioned in Section 3 and stored in the CSV file. In addition, the historical records of battery SoC, or varying distance, time and speed are collected as discussed in Section 3. The dataset is then pre-processed to remove NaN and missing values. The values are then normalized to bring down the feature range [0,1]. The processed data is finally stored in CSV file. The backend includes distance calculator based on EV latitude and longitude and geo-locations of all EVCS, stored in database. This deploys Haversine formula. The distance calculator is analysed for its accurate response for the source geo-location of **Amrita Vishwa Vidyapeetham, Ettimadai, Coimbatore, Tamil Nadu, India** (10.9085, 76.9098) and destination geo-location of **Zeon Charging Station, Navakkai, Coimbatore, Tamil Nadu, India** (10.85592, 76.86251). The source latitude is found to be 0.1904 rad and longitude is obtained as 0.1941 rad, while destination latitude is found to be 0.1895 rad and longitude is obtained as 1.3415 rad. The difference between latitudes is calculated using equation (4) and is found to be 0.0037 rad, while the difference between longitudes is calculated using equation (5) and is found to be -0.0009177 rad. Deploying (1) and (2), a is obtained as 3.748×10^{-7} and 0.0012244 respectively. Finally, the distance between the source and destination is computed and is observed as 27.65 km. The obtained values are verified through online calculators and the code for the calculation of the distance between the source and destination is incorporated in the backend. The code is tested for the source geo-location of (10.9085, 76.9098) and destination geo-location of (10.85592, 76.86251). The obtained result is found to match with the manual calculation. Post verifying, the code is modified to compute the distance between the current EV location and each EVCS location, available in the database. The details of EVCS, along with the distance, are sorted in ascending order of distance.

EVCS Locator

Find a Charging Station

Select your location to find nearby stations

Use My Current Location

Your coordinates: Latitude 10.9085, Longitude 76.9098

Select a state

OR
 Select a city

Find Stations

Nearby Stations

Adyar Ananda Bhavan Navakkarai
 Navakkarai, Tamil Nadu
 6.67 km away
 Operator: Zeon Charging
 Connector: CCS (Type 2)
 Power: 50kW
 Predicted SoC: 2.4%

Recalculate SoC

Sundaram Motors Coimbatore
 Coimbatore, Tamil Nadu
 9.48 km away
 Operator: Chargezone (India)
 Connector: CCS (Type 2)
 Power: 45kW
 Predicted SoC needed: 3.4%

Recalculate SoC

Velanthavalam
 Vellamthavalam, Kerala
 11.73 km away
 Operator: (unknown)
 Connector: CCS (Type 2)
 Power: 30kW
 Predicted SoC needed: 4.22%

Recalculate SoC

Ahalia, Walayar – GO EC
 Walayar, Kerala
 12.0 km away
 Operator: GO EC
 Connector: CCS (Type 2)
 Power: 30kW
 Predicted SoC needed: 4.32%

Recalculate SoC

Table 3. XGBoost and Random Forest Predicted and actual SoC.

Sample No.	Predicted Value Random Forest	Predicted Value XGBoost	Actual Value
1	56.06	56	56
2	70.43	70.29	69
3	84.90	84.92	85
4	90.68	90.85	91
5	70.35	69.36	68

Fig. 3. Front end – Proposed EVCS recommendation system.

In the next step, the history EV battery SoC dataset that is pre-processed is split to create train-test data. The 80% data is deployed to train the random forest with $n_estimators$ as 100, max_depth of 15 and $random_state$ as 42. While XGboost model is trained with $n_estimator$ of 300, learning rate of 0.05, max_depth of 6 and $random_state$ of 42. The models are tested with 20% data. During the performance evaluation

of random forest, the MSE, MAE, RMSE and R^2 score are obtained to be 28.6034, 1.9594, 5.3482 and 0.9304 respectively. Meanwhile, XGBoost provides MSE, MAE, RMSE and R^2 score of 27.7796, 1.9054, 5.2706 and 0.9324 respectively.

The bar chart representation illustrating the performance indices corresponding to the two models is depicted in Fig. 2. A sample of actual and predicted SoC by XGBoost model based on the distance, speed and time is tabulated in Table 3. The observed performance indices (Fig. 2) and predicted values (Table 3) show the slightly better performance of XGBoost in predicting SoC. Thus, XGBoost model is selected for the deployment of the proposed approach. The front end is developed using Next.js, React, and Tailwind CSS. The UI components are from the ShadCN UI library. The code is incorporated with built-in Geolocation API to pass the current EV Geolocations to the backend. Meanwhile, the backend and database are interfaced by an API layer that facilitates seamless communication between the two models. The API allows the backend to query and retrieve EVCS details stored in TypeScript array of objects, which is like JSON. The front end – backend – database interface allows the backend to compute the distance of EV with respect to all EVCS.

The backend also includes the trained SoC predictor model, which then computes the SoC post arriving at the nearest four destinations. The computed value is then pushed to the front end to notify the user. The front end of the proposed AI assistance is presented in Fig. 3. The details (shown in Fig. 3) of four nearest EVCS that include station name, city, state, operator name, type, connector type and power output are correctly fetched and displayed by the proposed system. The obtained result for the current EV location is analyzed and verified through online resources and real time tracking. The proposed model is analyses for different EV locations. To confirm the satisfactory response, the proposed model is tested for a new source latitude and longitude of 10.05 and 77.05. The given location corresponds to Pallivasal Tea Factory, Munnar, Kerala. The system recommends 4 nearest EVCS as Eden Wood Resorts, Eastend Hotel and Resorts, Anert EESL - DTPC Idukki Park and Calvarymount Jerry Plaza located away from the source by 6.90km, 7.84km, 23.716387km and 27.027704km respectively. The predicted SoC needed is observed as 0.984962, 1.565746, 8.537899 and 9.729973 respectively. The recommendation details are found to be acceptable while verifying using online resources.

A real-time evaluation and validation of EVCS recommender is conducted at different locations in the borders of Kerala – Tamil Nadu and Kerala due to adequate availability of EVCS details in the dataset and ease of real time testing. The observed results are found to be satisfactory. The results show the effective operation of the AI assist to recommend the nearest EVCS. The proposed computationally efficient, lightweight, intelligent assistance ensures a promising solution to real-time EV users, enhancing convenience, supporting efficient battery utilization and contributing towards the large-scale electric mobility.

5 Conclusions

This work presents an intelligent assistant framework that leverages geo-location, predicted SoC, dynamic transit EV characteristics to provide real-time EV charging station recommendations. The proposed system uniquely integrates latitude–longitude–based distance calculation with machine learning–driven SoC forecasting to deliver personalized, real-time recommendations on the top 4 EV charging station. Random Forest and XGBoost models that are modelled deploying the outcome of the Haversine-based distance calculator are compared using the performance indices. The performance comparison between the two ensemble learning models demonstrates that XGBoost marginally outperforms Random Forest in SoC prediction. Specifically, XGBoost achieves a low MAE of 1.9054, RMSE of 5.2706, MSE of 27.7796, and a slightly higher R^2 of 0.9324, compared to Random Forest model. This indicates more accurate and robust SoC estimation of XGBoost compared to Random Forest. The proposed system is analyzed using real-world test data across the Kerala–Tamil Nadu border. The results confirm responsiveness and reliability of the model in assisting EV travelers with decisions on charging station selection. The findings highlight the potential of AI-driven frameworks to improve usability, while alleviating range anxiety, and supporting the global transition towards sustainable mobility. Though the proposed approach provides a satisfactory result, a few limitations are identified. Factors like ambient temperature, road traffic, availability of charging station and driver behavior when fully integrated, could impact the practical accuracy. Model extension to account for variations in vehicle type, battery chemistry and battery aging and **addition of charging station details across different states**, can make the system more widely applicable. Future work can incorporate graph-based algorithms or route planning APIs to complement the Haversine distance, improving the accuracy of arrival-time predictions and energy consumption.

References

1. Sacchi, R., Bauer, C., Cox, B., Mutel, C.: When, Where and How Can the Electrification of Passenger Cars Reduce Greenhouse Gas Emissions?. *Renew Sustain Energy Rev.* 162, 112475 (2022).
2. Wuebbles, D. J., Jain, A. K.: Concerns about Climate Change and the Role of Fossil Fuel Use. *Fuel Process Technol.* 71(1), 99–119 (2001).
3. Chapman L.: Transport and Climate Change: A Review. *J Transp Geogr.* 15(5), 354–367 (2007).
4. Shingwun, Sun, Tian, Li: An Analysis of the Global EV Market: Sustainable Pathways to Net Zero Emissions. *Journal of Applied Economics and Policy Studies* 11, 67–80 (2024).
5. Jung, Felix , Schröder, Malte, Timme, Marc: Exponential adoption of battery electric cars. *PLoS One* 18(12), 0295692 (2023).
6. Swaminathan, Arihant, Nirmal, Ashwin, Kishore Krishna, L. S., Kruthik Soundappan, R., Lekshmi, R. R., Vijaya Chandrakala, K. R.M.: Development of Low-Cost Electronic Control Unit for Hybrid Electric Vehicle. *Proc. International Conference on Paradigm Shifts in*

- Communications Embedded Systems, Machine Learning and Signal Processing, IEEE, Nagpur, India, pp. 1 – 6 (2023).
7. Md. Shahadat Hossain, Mahmudur Rahman Fatmi, Mostaq Ahmed, Bijoy Saha: “Where to plug in? Assessing the users’ preferences for EV charging location. *International Journal of Transportation Science and Technology*, Preprint (2025).
 8. Boubaker, S., Al-Dahidi, S., Kamel, S., Ghazouani, N., Kraiem, H., Alsubaei, F.S., Bourennani, F., Meskine, W., Benghanem, M., Mellit, A.: Electric vehicles charging station allocation based on load profile forecasting and Dijkstra’s algorithm for optimal path planning. *Scientific Reports* 15(1), 23844 (2025).
 9. Shreyashree, P.M., Bhumika, A., Kodipalli, A., Rao, T., Suman, J.V.: Electric Vehicle Charging Stations: A Comparative Study of Multiple Machine Learning Classifiers and Interpretation Using Explainable AI. *Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering* 620, 321–331 (2026).
 10. Savari, G. F., Krishnasamy, V., Sathik, J., Ali, Z. M., Aleem, S. H. E. A.: Internet of Things based real-time electric vehicle load forecasting and charging station recommendation. *ISA Trans.* 97, 431–447 (2020).
 11. Wang, Jiyao, Zhang, Wenbo, Wen, Xiao Luke, Ran, Tu, Dengbo, He.: Modeling battery electric vehicle users’ charging decisions in scenarios with both time-related and distance-related anxiety. *Travel Behaviour and Society* 42, 101111 (2026).
 12. Lee, D., Kim, D.S., Chung, B.J., Chung, Y.M.: Recommendation of Electric Vehicle Charging Stations in Driving Situations Based on a Preference Objective Function. *World Electric Vehicle Journal* 16(4), 192 (2025).
 13. Sun, B., Li, T., Low, S. H., Tsang, D. H. K.: ORC: An online competitive algorithm for recommendation and charging schedule in electric vehicle charging network. *Proc. 11th ACM Int. Conf. Future Energy Syst.*, pp. 144–155 New York, NY, USA: 2020 (2020).
 14. Xiong, J., Gu, H.: An optimal method for charging station location based on big data of the SOC and immune algorithm. In *Proc. Smart World, Ubiquitous Intelligence & Computing, Advanced & Trusted Computed, Scalable Computing & Communications, Cloud & Big Data Computing, Internet of People and Smart City Innovation*, pp. 1 – 7 IEEE, San Francisco, CA, USA, (2017).
 15. Peidong, Xu, Jun, Zhang, Tianlu, Gao, Siyuan, Chen, Xiaohui, Wang, Huaiguang, Jiang, Wenzhong, Gao: Real-time fast charging station recommendation for electric vehicles in coupled power-transportation networks: A graph reinforcement learning method. *International Journal of Electrical Power & Energy Systems* 141, 108030 (2022).
 16. Li, H., Han, Q., Bai, X., Zhang, L., Wang, W., Chen, W., Xiang, L.: Electric Vehicle Charging Station Recommendations Considering User Charging Preferences Based on Comment Data. *Energies* 17: 5514 (2024).
 17. Xiaohui, Wang, Xiaokun, Zheng, Xiao, Liang: Charging Station Recommendation for Electric Vehicle Based on Federated Learning. *Journal of Physics: Conference Series* 1792, 012055 (2021).
 18. Vishal Chaganti, P. C., Manitha, P. V., Sailaja, V.: Nearest Charging Station Identification for EV Using Machine Learning Techniques. In *3rd International Conference on Emerging Frontiers in Electrical and Electronic Technologies*, pp. 1 – 6 IEEE, Patna, India, (2023).
 19. Sreerag, S., Lekshmi, R. R.: Allocation of Medicine Dispensing Mobile Robot Considering Path Plan and Battery SoC. In *Proc. International Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation*, pp. 1 – 6 IEEE, Gwalior, (2024).
 20. Sreekumar, A. V., Lekshmi, R. R.: Comparative Study of Data Driven Methods for State of Charge Estimation of Li-ion Battery.: In *proc. International Conference on Paradigm*

- Shifts in Communications Embedded Systems, Machine Learning and Signal Processing, pp. 1 – 6 IEEE, Nagpur, India, (2023).
21. Desai, J., Mathew, J.K., Sturdevant, N.J., Bullock, D.M.: Longitudinal Monitoring of Electric Vehicle Travel Trends Using Connected Vehicle Data. *World Electric Vehicle Journal* 15(12), 560 (2024).
 22. Rezanía Agramanisti Azdy, Febriyanti Darnis: Use of Haversine Formula in Finding Distance Between Temporary Shelter and Waste End Processing Sites. *J. Phys.: Conf. Ser.* 1500 012104 (2020).
 23. Prajwol Sangat, Maria Indrawan-Santiago, David Taniar, Beng Oh, and Paul Reichl: Processing High-Volume Geospatial Data: A case of monitoring Heavy Haul Railway operations. *Procedia Computer Science* 80, 2221–2225 (2016).
 24. Anil Sahebrao Kapile, Ramesh, D. Shelke, and Nilesh, B. Kalani: SOC Estimation of E-Vehicle Using Large Dataset for Vehicle Energy Consumption. *Journal of Harbin Engineering University* 44(10), 1291–1299 (2023).
 25. <https://www.kaggle.com/datasets/pranjal9091/ev-charging-stations-in-india-simplified-2025>
 26. <https://www.kaggle.com/datasets/officialdatasets/battery-soc-dataset>
 27. Lekshmi, R. R., Annirudh, D., Surya, R., Kousik Harish, S., Srilekha, S.: Hardware Prototype Model of Conventional Gas Stove Automation System – Application of Random Forest Regression Algorithm. *Lecture Notes in Electrical Engineering*, Springer, Singapore, 977, 1005–1016 (2023).
 28. Bhardwaj, A., R. D., Devi, V. S. K.: An Enhanced Framework for Churn Prediction using Stratified Bagging and Stacked Multi-Output Random Forests. In *proc. International Conference on Intelligent Data Communication Technologies and Internet of Things*, pp. 119–123 IEEE, Bengaluru, India (2025).
 29. Maitra, P. S., Remya, R., Singh, A.: Groundwater Variability Along with the Varying Coastline of Thiruvananthapuram City. *J. Eco Res & Rev* 3(4), 298–303 (2023).